



Online Shopping Cart System

C++ — Case Study No. 170

Abdeali Makda

Roll No: 150096725096 | Cohort: Larry Page
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Background

Academic C++ assignments often focus on isolated concepts without showing how they combine into a complete system. Students struggle to connect abstract OOP theory to real-world applications.

E-commerce — one of the most ubiquitous software domains — offers an ideal context: shopping carts, billing engines, and order histories are patterns every developer will encounter.

Core Challenge

Case study focuses on developing an Online Shopping Cart System using C++. The system simulates the core functionality of an e-commerce platform by allowing users to add products to a cart, remove items, view total cost, and generate a final bill during program execution.

Key Requirements

01

Product Details

ID, Name, Price per item

02

Shopping Operations

Add, Remove, View Cart

03

Billing Engine

GST calc + discount support

04

Order Persistence

Save/retrieve order history

05

OOP Architecture

8 C++ concepts demonstrated

OOP Architecture

Abstract Product base class with Electronics & Accessory derived classes. Full polymorphism via virtual functions.

Cart Management

ShoppingCart manages up to 20 CartItems with add, remove, duplicate-check (findIndex), and array compaction.

GST Billing

18% GST computed per unit via virtual getGST(). viewCart() renders formatted table with base, GST & subtotal.

Discount Engine

Checkout accepts 0–100% discount with validation, computing final payable amount with full bill breakdown.

File Persistence

saveOrder() appends to orders.txt using ofstream (ios::app) in pipe-separated format. Retrieval via ifstream.

Menu Interface

do-while loop with switch-case for 6 options. Edge cases (empty cart, invalid input) handled gracefully.

10

Products

2

Categories

18%

GST Rate

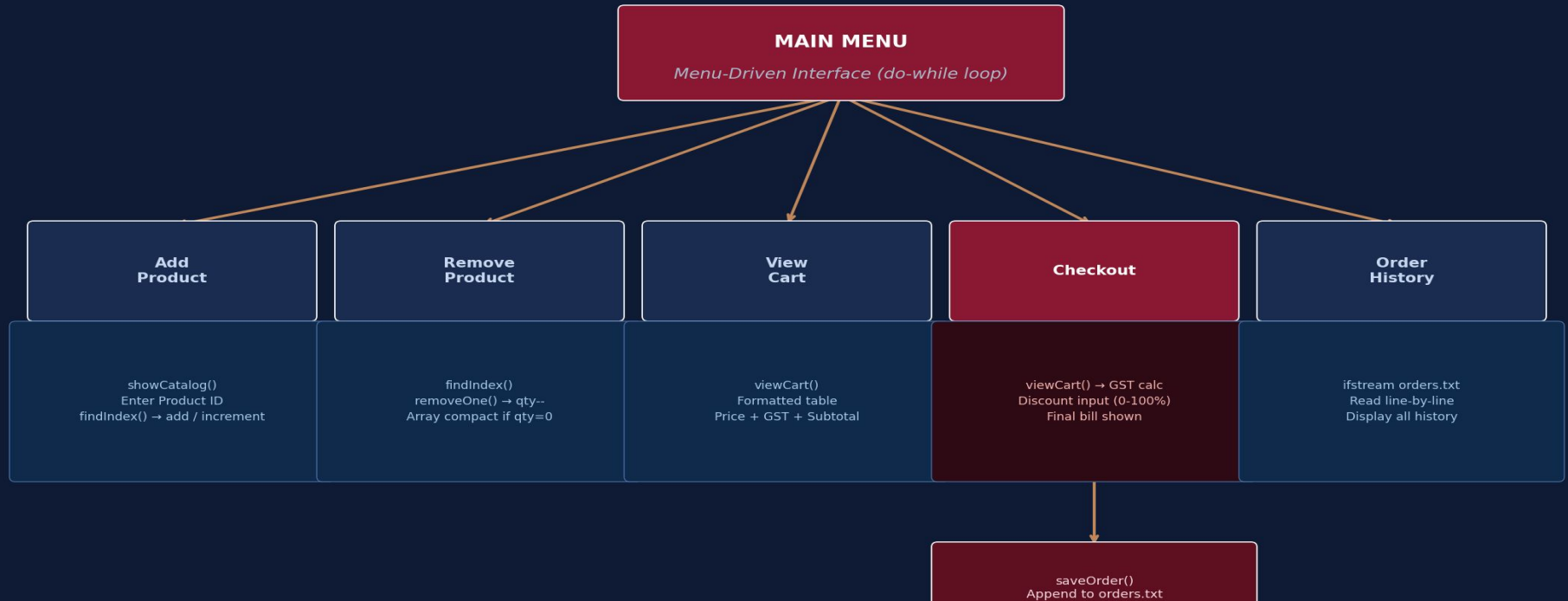
8

OOP Concepts

1

Source File

Online Shopping Cart System — Program Flow



What Was Achieved

The Online Shopping Cart System successfully demonstrates all 8 targeted C++ and OOP concepts — classes, abstraction, inheritance, polymorphism, encapsulation, virtual destructors, dynamic memory, and file handling — in a single, coherent application.

The clean separation between the Product hierarchy, CartItem, and ShoppingCart mirrors the single-responsibility and open-closed principles found in professional software design.

Current Limitations

No persistent cart state • Fixed-size arrays (MAX 20 items) • Single GST rate (18% for all)
No user authentication • Single-user, single-session model

Future Scope

- Replace arrays with `std::vector` for dynamic sizing
- Category-specific GST rates (multiple slabs)
- User authentication & multi-session support
- Coupon/promo code system
- Graphical or web-based frontend (REST API)
- Persistent cart state between sessions



Thank You

Abdeali Makda | 150096725096 | Cohort: Larry Page
School of Future Tech | ITM Skills University